Methods | MTH1002 | Term 2
Dr. Mike O'Sullivan | Year: 2025-2026

Exercise Sheet 3 (Tutorials in Week 6)

Key:

- **Questions marked with a ☕** are for discussion in your tutorials.
- **Questions marked with a 🐦‍⬛** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

☕ **Exercise 1.** If $u = u(x, y)$ and $x = e^s \cos t, y = e^s \sin t$, show that

$$u_{xx} + u_{yy} = e^{-2s} \left(u_{ss} + u_{tt} \right).$$

☕ **Exercise 2.** Find $\partial z/\partial x$ and $\partial z/\partial y$ if the equations

$$(i)\ z^3 - 3xyz = a^2, \quad (ii)\ x + y - z = e^z$$

define z as a function of the two independent variables x and y.

☕ **Exercise 3.** Find the Taylor series expansion for $f(x, y) = \sin x \cos y$ about the point $x_0 = \pi/4$, $y_0 = \pi/4$, up to and including the quadratic terms.

🎵 **Exercise 4.** Find and classify all the critical points of the function

$$f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 - 2.$$

🎵 **Exercise 5.** Use the Lagrange multiplier method to find the maximum and minimum values of the function $F(x, y) = x^3 y^5$ on the ellipse

$$3x^2 + y^2 = 8.$$

🎵 **Exercise 6.** Find the point on the paraboloid $z = x^2 + y^2$ which is closest to the point $(3, 3, 1)$. [Hint: The distance from (x, y, z) to $(3, 3, 1)$ is $\sqrt{(x-3)^2 + (y-3)^2 + (z-1)^2}$. Minimize the *square* of this distance for (x, y, z) on the paraboloid.]

♫ **Exercise 7.** Find the maximum value of the function $f(x, y, z) = \ln x + \ln y + 3 \ln z$ on the sphere

$$x^2 + y^2 + z^2 = 5R^2, \quad x > 0, y > 0, z > 0.$$

Hence prove that

$$abc^3 \leq 27 \left(\frac{a+b+c}{5} \right)^5$$

for any positive numbers a, b, c. Hint: Consider

$$F(x, y, z, \lambda) = \ln x + \ln y + 3 \ln z + \lambda(x^2 + y^2 + z^2 - 5R^2).$$